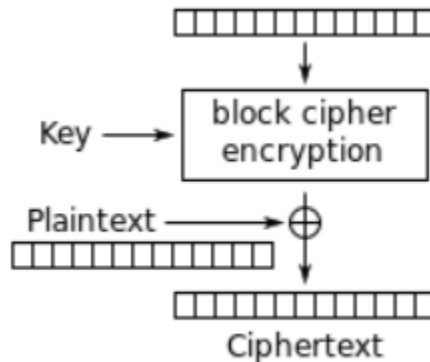


REPORT

The AES (Advanced Encryption Standard) Counter (CTR) mode is a block cipher that creates a pseudorandom keystream by encrypting a counter value with the encryption key. The ciphertext is then created by XORing this keystream with the plaintext.

A counter is initialized in CTR mode with a distinct value.



To create a block of keystream, the counter is encrypted with the AES key.

To create the ciphertext, the keystream and the matching block of plaintext are XORed. After that, the procedure is repeated with the counter increased for the subsequent block.

Initially for finding the key we can do a brute force attack, this attack will work for 16 bit but not for 32,48 bit so for that we need to find new type of attack by using the properties of AES.

Attack Using Chosen Plaintext in AES CTR Mode: In this type of attack, We have the freedom to select and encrypt any plaintext block they desire.

We can take advantage of this predictability if we are able to predict the counter values that are utilized during encryption.

We can execute XOR operations between the corresponding ciphertext blocks and the known plaintext if they are aware of specific portions of the plaintext. A portion of the keystream is revealed by this XOR operation.

Keystream Recovery: Additional portions of the keystream can be inferred using the partial knowledge about the keystream that was received by the XOR operation with the known plaintext. The keystream for every block is generated by the attacker by taking advantage of the encryption of the same counter value.

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[illegible]

Recovering Entire Plaintext:

With a sufficiently large amount of known plaintext and corresponding ciphertext, the attacker can recover the entire keystream. Once the keystream is obtained, it can be XORed with the remaining ciphertext to reveal the entire plaintext message.

Implications:

A successful chosen plaintext attack on AES CTR mode can compromise the confidentiality of the data. We can recover the plaintext without knowing the actual encryption key.